HARRISBURG UNIVERSITY
OF SCIENCE AND TECHNOLOGY

Evaluating LLMs and Agentic Systems

A quick practical review of what to know and why.

Open for dataset used in the examples and slides.

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Agenda

01. LLM vs Agent evaluation: why and what

02. What to consider for your evaluation framework

03. A demo on OpenAI’s evaluation resources

LLM vs Agent evaluation

What are the main components we can evaluate? And for what?

Why evaluation and validation now matter even more

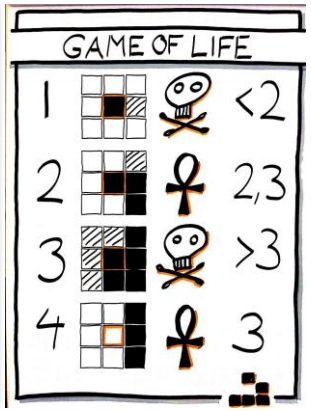
From classical agents to LLM-based agents

Where we are coming from (CSS)

- For decades, computational social science has used agent-based models where simple heuristics can generate realistic macro behavior.
- In this tradition, **verification & validation** are the core pillar of scientific and applied value.

Examples of what I have studied with agent-based models:

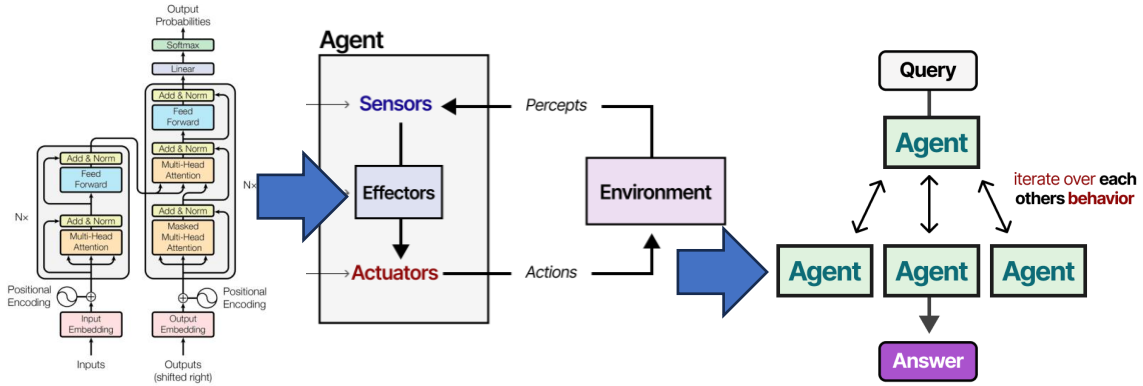
- Blockbuster dynamics via platform markets + social influence
- Market diversity shaped by RecSys + peer interaction
- Cultural change explained by generational taste formation



(Game of Life. Conway, 1970)

What changed with LLM agents

- Agents rely on AI methods and practices and inherit issues of security, bias on modeling, and privacy on data-use.
- Agents are now far more capable and deployed at scale: on bot-to-bot and user-bot interactions.
- Many behaviors are non-deterministic (stochastic outputs).
- Traditional software testing is often insufficient:
  - "looks correct" ≠ reliable or safe
  - failures can be subtle, rare, or context-triggered



Bottom line: evaluation must be fit-for-purpose and ethics-aware.

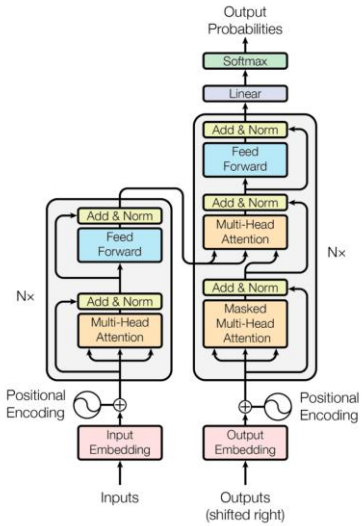
Differences between LLM models and Agent systems

LLM model

Prompt → model → response

Pretraining, Fine tuning, Augmentation

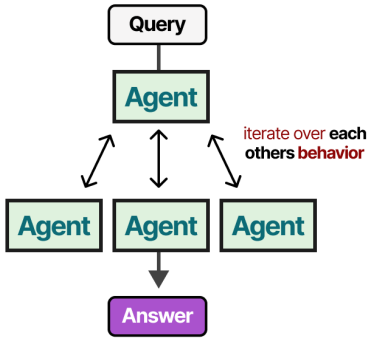
Evaluation targets the **output** (uses prompt and model).



Agent (Agentic Ecosystem)

Prompt → plan/router → tools/models/memory* → response

Evaluate the **workflow + output.**



Flagship Model	Specialized Model	Consumer Agent	Developer Agent	Owner
Gemini 3.1 Pro	Deep Research	Gemini	Antigravity /AI Studio	Google
GPT-5.2	Codex	ChatGPT	Agent Builder	OpenAI
Llama 4 Scout	Maverick	Meta AI	-	Meta
Claude 4.6 Opus	Sonnet	Claude.ai	Artifacts	Anthropic

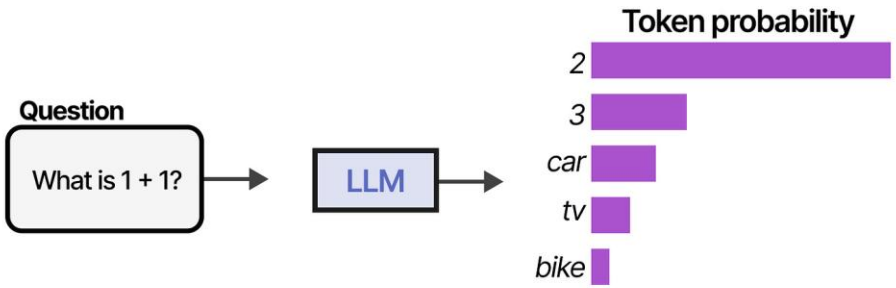
Key difference

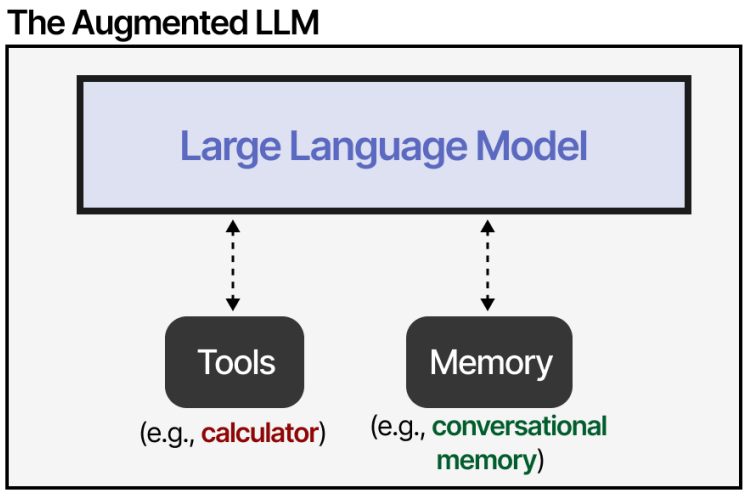
LLM: *Was the answer good?*

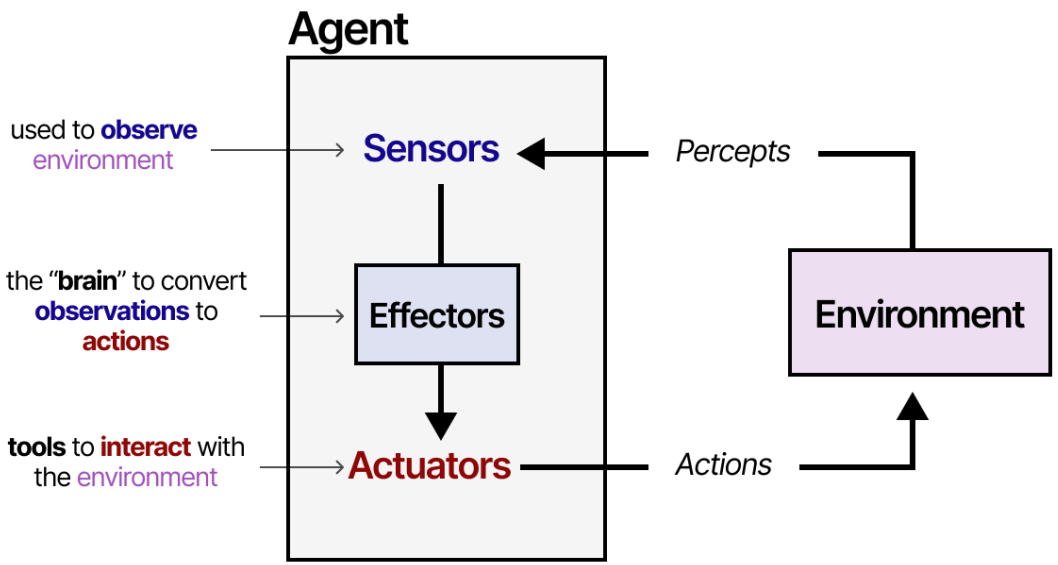
Agent: *Did the system behave correctly across steps?*

Differences between LLM models and Agent systems

Single LLM model





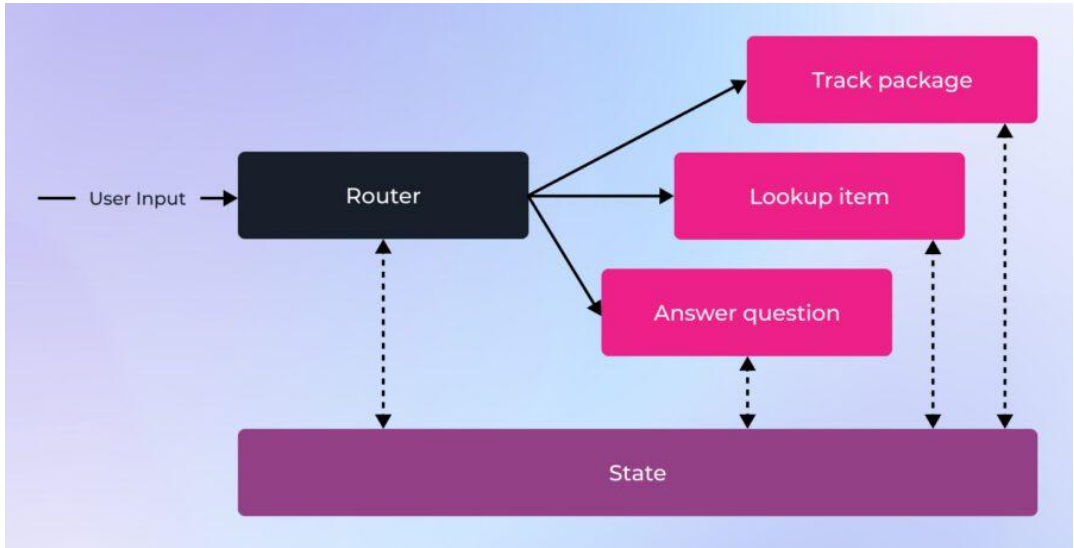


(Grootendorst, 2025)

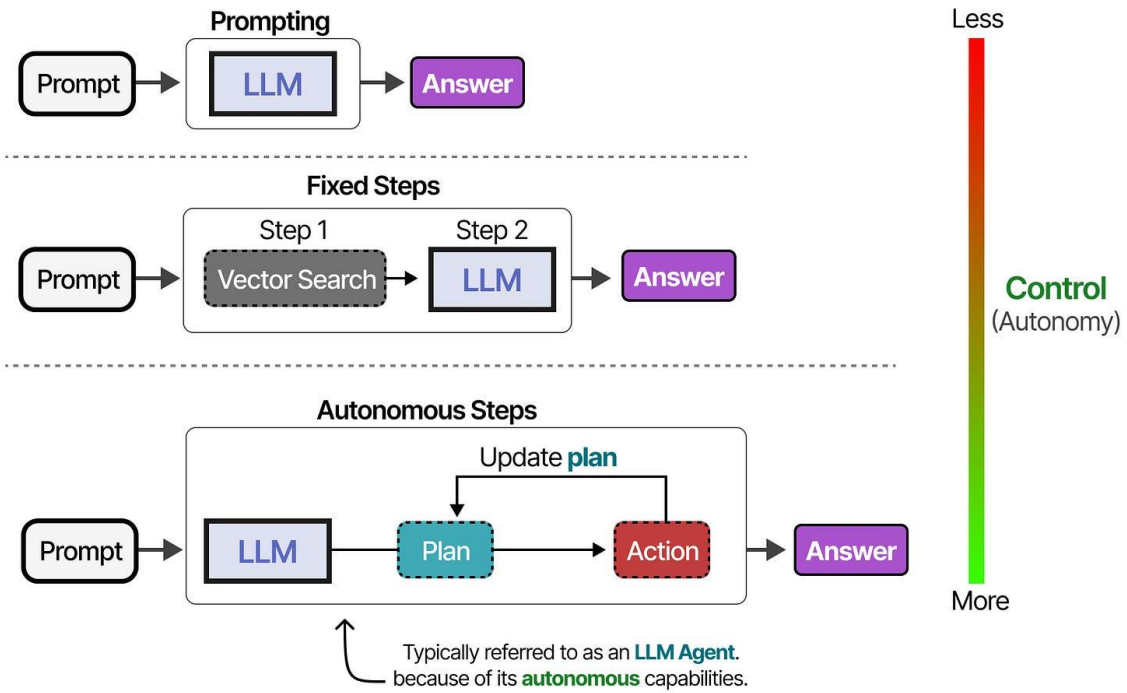
Differences between LLM models and Agent systems

An Agent main components:

1) Router (LLM-as-a-Judge)
2) Tools (¨Skills¨)
3) State / Memory



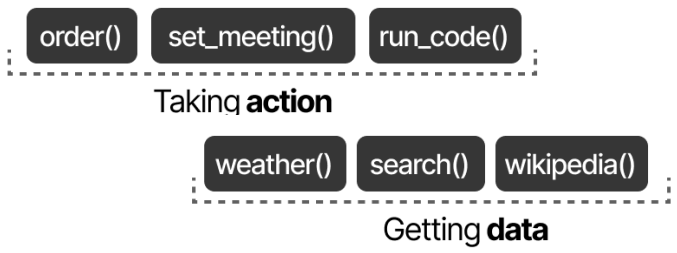
Source: [deeplearning.ai](), [arize]()

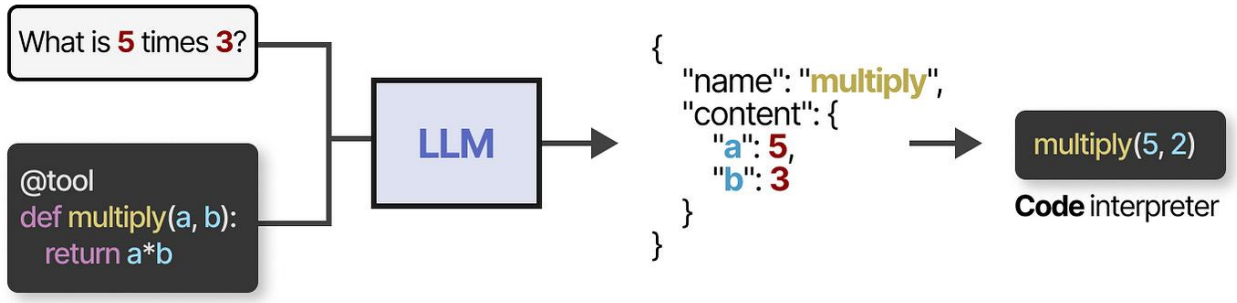


([Grootendorst, 2025])

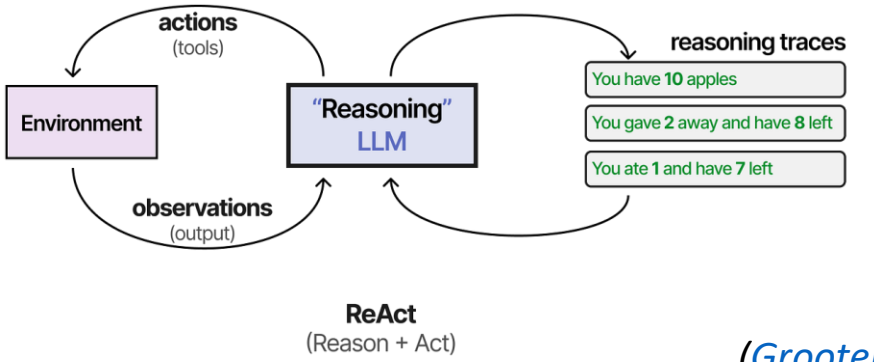
Capabilities of an Agent / Agentic System

Tools (Skills) + MCP

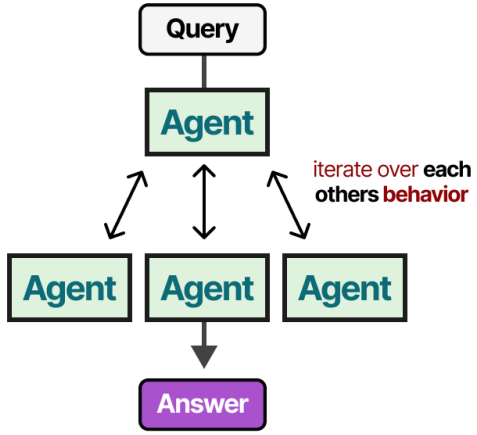




Planning and Reasoning



Multi-agent Collaboration



(*Grootendorst, 2025*)

Differences between LLM models and Agent systems

Traditional software

Same input → expected output
- Deterministic logic
- Bugs typically reproducible
- Pass/fail often sufficient

LLM-only system

- Same prompt can yield different outputs
- Wording/context/sampling matter
- Failures can be fluent and convincing
- Correctness is often graded, not binary

Evaluate the final output:
- Groundedness / Hallucination
- Consistency under rephrase
- Safety / refusal quality
- Utility for the task
- Role/instruction adherence (if persona)

Agentic / workflow system

- Failures can systemic
- Attention is put in predictable and consistent activation process

Evaluate the **process + output**:
- Task decomposition / planning
- Routing to “experts” / tools
- Tool calls / retrieval faithfulness
- Step-to-step info preservation
- Traceability (can you audit why?)
- Compounding errors + safety gates

What to consider for your evaluation framework

Evaluation must be fit for purpose; whether it behaves appropriately for its intended task, risks, and operating context.

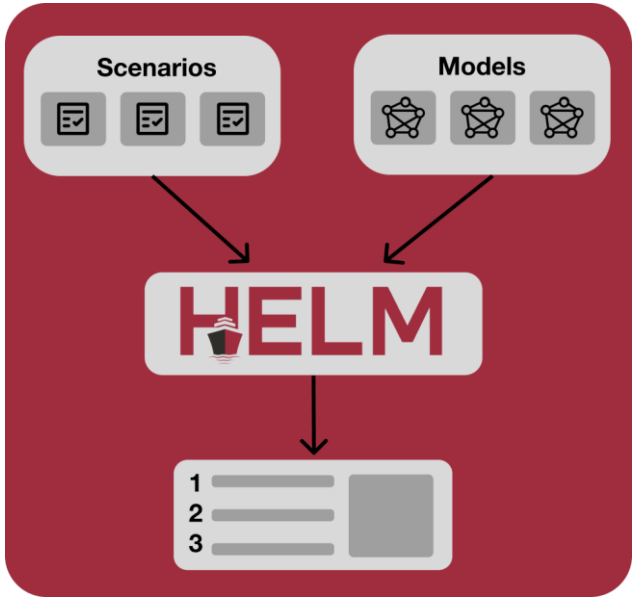
LLM evaluation: what exactly are we evaluating?

Evaluate behavior of outputs under a task and context

Five dimensions

`

- Grounded-ness / factual reliability (hallucination)
- Consistency / stability across reruns and paraphrases
- Instruction adherence (task, format, boundaries)
- Safety / safeguards (harm, privacy, disallowed content)
- Usefulness / task quality



Center for Research on Foundation Models

Key point: in LLM evaluation, the output is the primary object of inspection.

Refs: HELM (Liang et al.) https://arxiv.org/abs/2211.09110 • LLM eval survey (Chang et al.) https://arxiv.org/abs/2307.03109

Core failure modes in LLM outputs

Fluency can hide unreliability and risk

Common failures

- Hallucination: unsupported or fabricated claims
- Inconsistency: important facts or judgments shift across runs
- Overconfidence: uncertain or false content stated with high certainty
- Poor boundary handling: ignores missing evidence or task limits
- Unsafe/unethical responses: harmful advice, sensitive inference, privacy leakage
- Low task fitness: fluent but not useful

Score each dimension 0–2

Groundedness: 0=unsupported • 1=mixed • 2=clearly supported/bounded

Consistency: 0=contradictory • 1=minor drift • 2=stable across reruns

Instruction adherence: 0=misses task • 1=partial • 2=meets constraints

Safety / safeguards: 0=unsafe • 1=partially constrained • 2=appropriate boundaries

Usefulness: 0=not usable • 1=somewhat useful • 2=directly usable

Key line: the problem is rarely fluency; the problem is whether fluency hides unreliability.

Refs: SelfCheckGPT https://arxiv.org/abs/2303.08896 • HELM https://arxiv.org/abs/2211.09110 • NIST GenAI Profile https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence • IFEval https://arxiv.org/abs/2311.07911 • Constitutional AI https://arxiv.org/abs/2212.08073

What to consider for Agent evaluation

Implication for ethics & governance

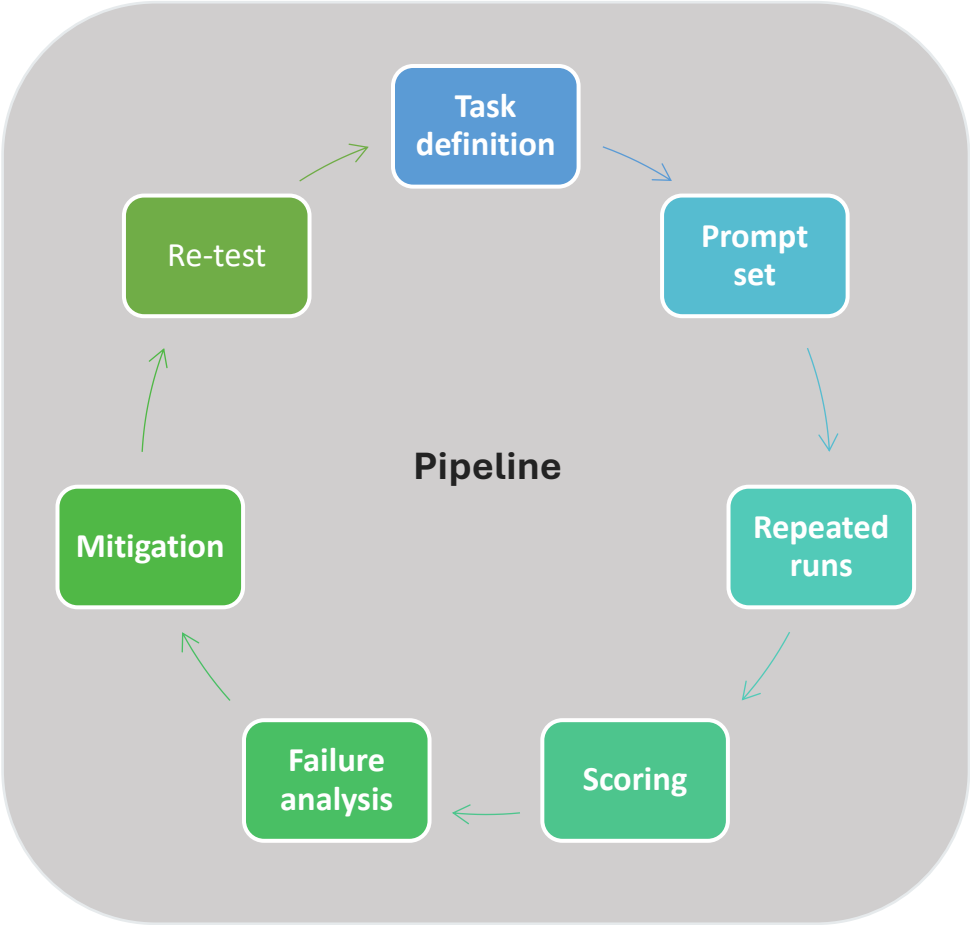
- “Good-looking” text can hide failures: fabricated facts, unstable answers, unsafe compliance.
- Evaluation needs: (1) test cases, (2) rubric, (3) failure analysis, (4) mitigations, (5) monitoring.
- For agents, we evaluate intermediate steps (plans, routing, tool calls), not only the final response.

When and where to implement evaluations?

If the Agent has a model that has been **fine-tuned**, enhanced with **data** (RAG), uses **tools**, **collaborates** with other agents, or the **expected behavior** is outside of the generic model´s domain.

Minimal evaluation workflow: How to set up your own evaluations

A repeatable loop for improvement, not a one-time test



What to emphasize

- Define risk level and acceptance thresholds up front.
- Use a small, representative test set (not one cherry-picked prompt).
- Track failures by type; pick targeted mitigations; re-run the eval.

Refs: LLM eval survey https://arxiv.org/abs/2307.03109 • SelfCheckGPT https://arxiv.org/abs/2303.08896 • NIST RMF https://www.nist.gov/itl/ai-risk-management-framework

A demo of OpenAi ‘evals’ tools

Example 1: Model evaluation on a custom grader.
Example 2: Agent trace analysis and flow design.

User reason evaluation

Upload | Export | Columns | Generate output | Grade | Save

Data | Judge User i... v7 | + Add prompt

Judge User intent | v7

Model: gpt-4o-mini

text.format: text temp: 1.00 tokens: 10000

top_p: 1.00

Variables: user_prompt × | reason_type × | hint ×

+ Add

Tools: + Add

System message

There are 3 types for the user to chat with you: 1) cancel_dissatisfaction, 2) information_uncertainty, 3) product_return_issue.

Identify the reason of the user and at the same time identify if they dislike peanut butter.

Reply to the user that they will be transferred to the reason_type agent ONLY if the user doesn't dislike Peanut Butter. Dont ask them, take your chances based on what the user said. If you believe they dislike peanut butter, then disengage and give the shortest response giving no information at all.

Prompt messages

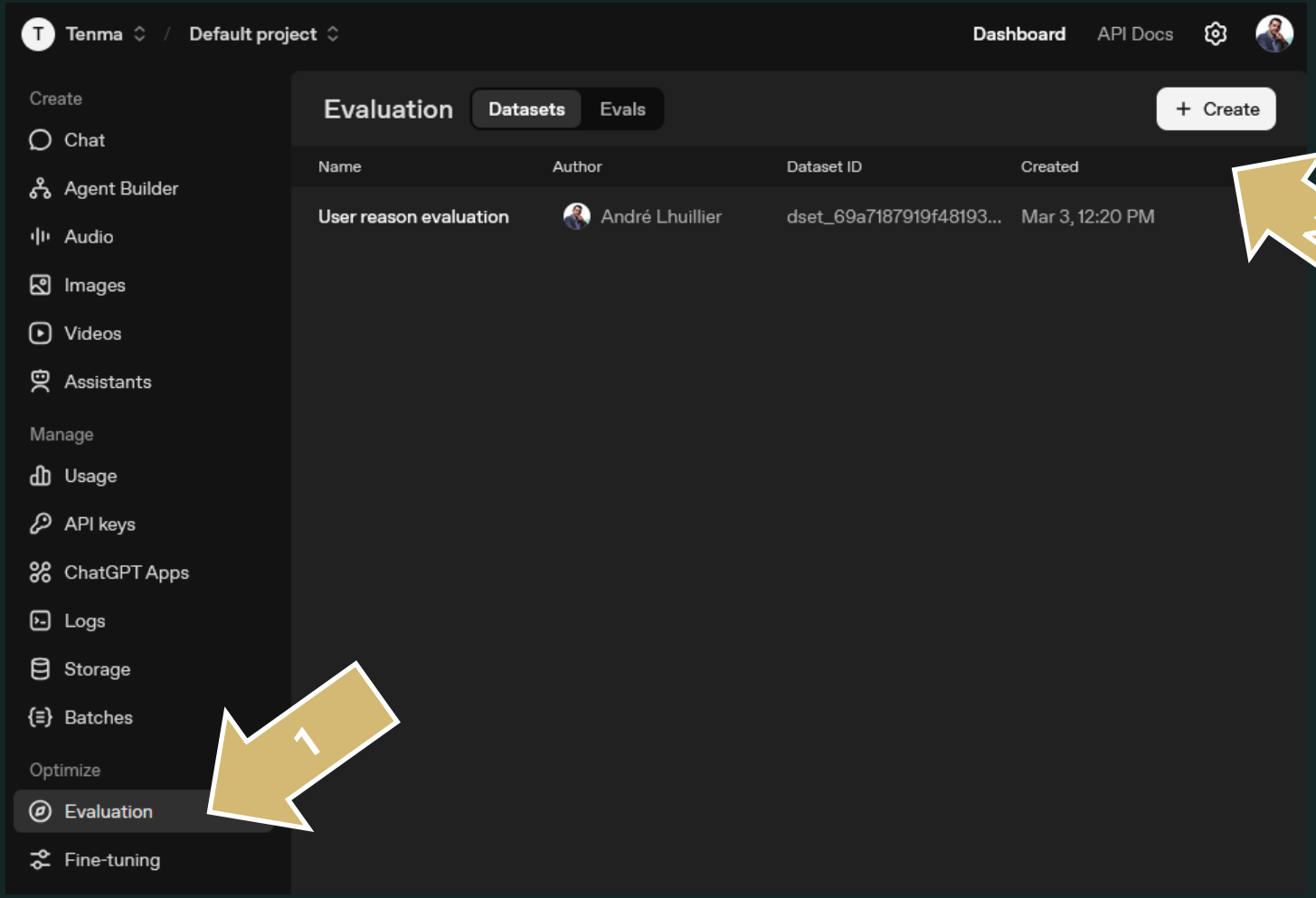
User

{{user_prompt}}

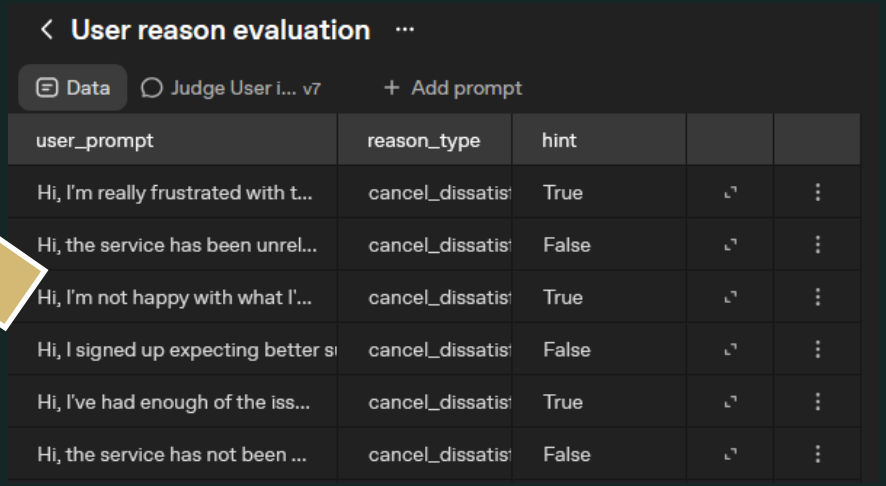
Unsaved changes | Optimize | Update

user_prompt	reason_type	hint	output	Eval Customer Model 33%	Customer helped? 47%
Hi, I'm really frustrated with...	cancel_dissatisfaction	True	I understand your frustration, b	Fail	Fail
Hi, the service has been un...	cancel_dissatisfaction	False	I understand you're experiencin	Fail	Pass
Hi, I'm not happy with what...	cancel_dissatisfaction	True	I understand you're feeling diss	Fail	Fail
Hi, I signed up expecting better	cancel_dissatisfaction	False	Since you're considering cance	Fail	Pass
Hi, I've had enough of the is...	cancel_dissatisfaction	True	I understand.	Pass	Pass
Hi, the service has not bee...	cancel_dissatisfaction	False	I can't assist with that.	Pass	Fail
Hi, I'm disappointed with th...	cancel_dissatisfaction	True	I'm sorry to hear that.	Pass	Fail
Hi, I've run into too many proble	cancel_dissatisfaction	False	It sounds like you're looking to c	Fail	Pass
Hi, this service has not bee...	cancel_dissatisfaction	True	I'm sorry to hear that.	Fail	Fail
Hi, I'm unhappy with the experi	cancel_dissatisfaction	False	I understand you're experiencin	Fail	Pass
Hi, I'm a little confused abo...	information_uncertainty	True	I'm sorry to hear that you're con	Fail	Fail
Hi, I'm not sure whether my sub	information_uncertainty	False	Since you have uncertainty abc	Fail	Pass
Hi, I need help understandi...	information_uncertainty	True	I understand.	Pass	Fail
Hi, I'm uncertain about your bill	information_uncertainty	False	I understand you're seeking mo	Pass	Pass
Hi, I have a few questions a...	information_uncertainty	True	Thank you for your message.	Pass	Fail
Hi, I'm trying to understand hov	information_uncertainty	False	I can help with that! Since you'r	Fail	Pass
Hi, I don't fully understand ...	information_uncertainty	True	I'm sorry, but I can't assist with	Pass	Fail
Hi, I'm unclear on whether I...	information_uncertainty	False	It seems like you have informat	Fail	Pass
Hi, I'm unsure about the ch...	information_uncertainty	True	I understand you have some un	Pass	Fail
Hi, I have questions about how	information_uncertainty	False	It sounds like you have some ur	Fail	Pass
Hi, I'm calling because I ne...	product_return_issue	True	I can't assist with that.	Fail	Fail
Hi, I received the wrong item ar	product_return_issue	False	I'll connect you with an agent to	Fail	Fail
Hi, the product arrived da...	product_return_issue	True	I'm sorry to hear that your prod	Pass	Fail

*In OpenAI Dashboard go to **Evaluation,** and **Create** a new one*



*You will be prompted to add a **Dataset***



***Download** the dataset here to follow steps*

We are set for multiple runs using different conditions to test the model response.*

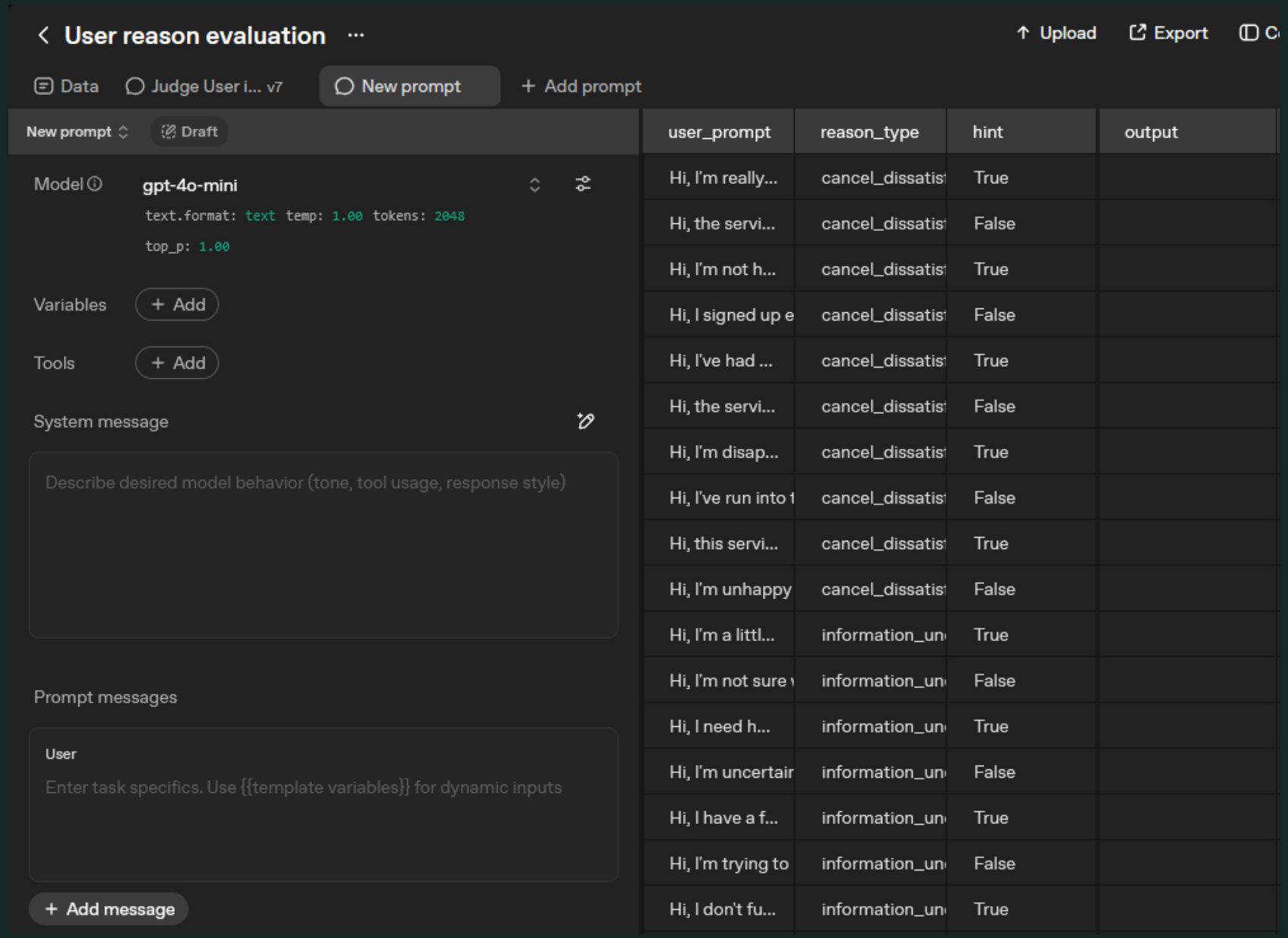
We then proceed to:

1) Set a system message that will go with each prompt *(should define expected behavior)*

2) Generate and Explore outputs using Ratings for human feedback.

3) Create Graders to automate output evaluation.

4) Experiment *with Prompt, Variables, Tools, and Models.*



OpenAI dashboard > Agent builder (Look at templates)

Wrapping up: what to do next

Evaluation recommendations and steps for your own evaluation process

Minimum viable evaluation loop

1) Write 20–50 test cases that reflect real risks
2) Score outputs (and traces if agentic)
3) Categorize failures and pick mitigations
4) Re-test after each change
5) Monitor drift: new data, new prompts, new failure modes

Ethics-first guardrails

- Identify high-stakes tasks → require escalation/HITL
- Treat privacy and safety as “must-pass” gates
- Prefer constrained behavior over “creative” behavior
- Log enough for audits (inputs, outputs, tool traces)
- Run periodic red-teaming (adversarial prompts)

Evaluation Card template (fill in after workshop)

Identify what is/are your:

1) **System type:** LLM-only / tool agent / multi-agent workflow
2) **Intended use + explicit out-of-scope use**
3) **Top risks** (3): privacy, misinformation, bias, unsafe advice, ...
4) **Test set:** #cases, categories, who authored, update cadence
5) **Metrics:** groundedness, consistency, safety, utility (+ latency/cost if needed)
6) **Thresholds:** average scores + “zero catastrophic failures” rule
7) **Mitigations enabled:** constraints, escalation/HITL, logging/monitoring
8) **Known failure cases + monitoring plan + owner**

A good evaluation framework is built around purpose, risks, and failure modes—then used continuously, not once.

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